Centre for Foundation Studies in Science
Universiti Malaya
Session 2025/2026 Semester 2
FAC1003 Tutorial 10

In this tutorial, we will practise on:
1)Function with argument and return value (non-void)
2)function with argument and no return value (void)

Answer ALL the questions.

1. In physics, an object in motion is said to have kinetic energy. The following formula can be used to determine a moving object's kinetic energy:

$$KE = \frac{1}{2}mv^2$$

where KE is the kinetic energy in Joules, m is the object's mass in kilograms, and v is the object's velocity in meters per second. Write a function that accepts an object's mass and velocity as arguments. The function should return the amount of kinetic energy that the object has. (Use function with argument and no return value (void))

2. Write a program in C++ to find the sum of the series $\frac{1!}{1} + \frac{2!}{2} + \frac{3!}{3} + \cdots \frac{n!}{n}$ using the function. (use function with argument and return value (non-void))

3. Suppose a particle is moving along the x-axis so that its position at time t is given by the formula:

$$f(t) = 3t^2 + 8t - 2t^{\frac{5}{2}}$$

(a) Write its velocity and acceleration as functions of t.
(b) Write a program demonstrating the functions to compute the particle position, velocity, and acceleration for time t = 0 until t = 1.5 with an increment of 0.1. (t = 0, 0.1, 0.2, … ,1.5).

Positive velocity refers to the particle moving in the positive direction (toward the right). Negative acceleration means that the velocity is decreasing at that point. (use function with argument and no return value (void))

4. Write a program demonstrating the functions to compute the average and standard deviation of four scores. Imply the sentinel value to end the program, as shown in the sample output file. The following formula can be used to calculate the average and standard deviation

$$\text{Average}, \bar{X} = \frac{\sum_{i=1}^{n} x_i}{n}$$

$$\text{Standard Deviation}, \sigma = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{X})^2}{n-1}}$$

The sample output file is shown below:

Press Y to enter the program or press any button to exit: Y Enter 4 numbers, I will give you the mean and standard deviation of the data.

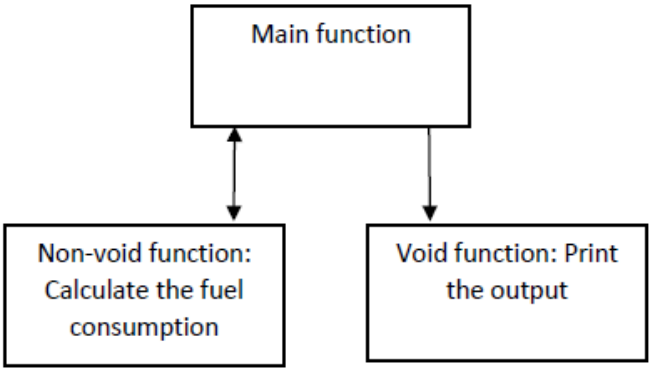
12.3
13.4
10.5
9

The mean of 12.30, 13.40, 10.50, and 9.00 is 11.30
The standard deviation of these numbers is 1.122497

Press Y to enter the program or press any button to exit: X You wished to exit. Thank you.

(use function with argument and return value (non-void))

5. The fuel consumption of a sedan car is 6.8 liters of petrol per 102 km. Write a program that displays a table showing the total fuel consumption of the sedan car for every 10, 20, 30, …, 150km. Your code must demonstrate the use of functions based on the following hierarchy.



The velocity (in m/s) of an object is given as a function of time (in seconds) by the following equation: $v(t) = 8t + 6t^2 - 16t^3$

Write a C++ program that prints an output of instantaneous acceleration of the object at seconds with an increment of 0.1. The program must include two functions named kiraPecutan and paparJadual. The kiraPecutan is a non-void function that calculates the instantaneous acceleration of the object, while the paparJadual is a void function for printing the desired output. (use multiple functions)